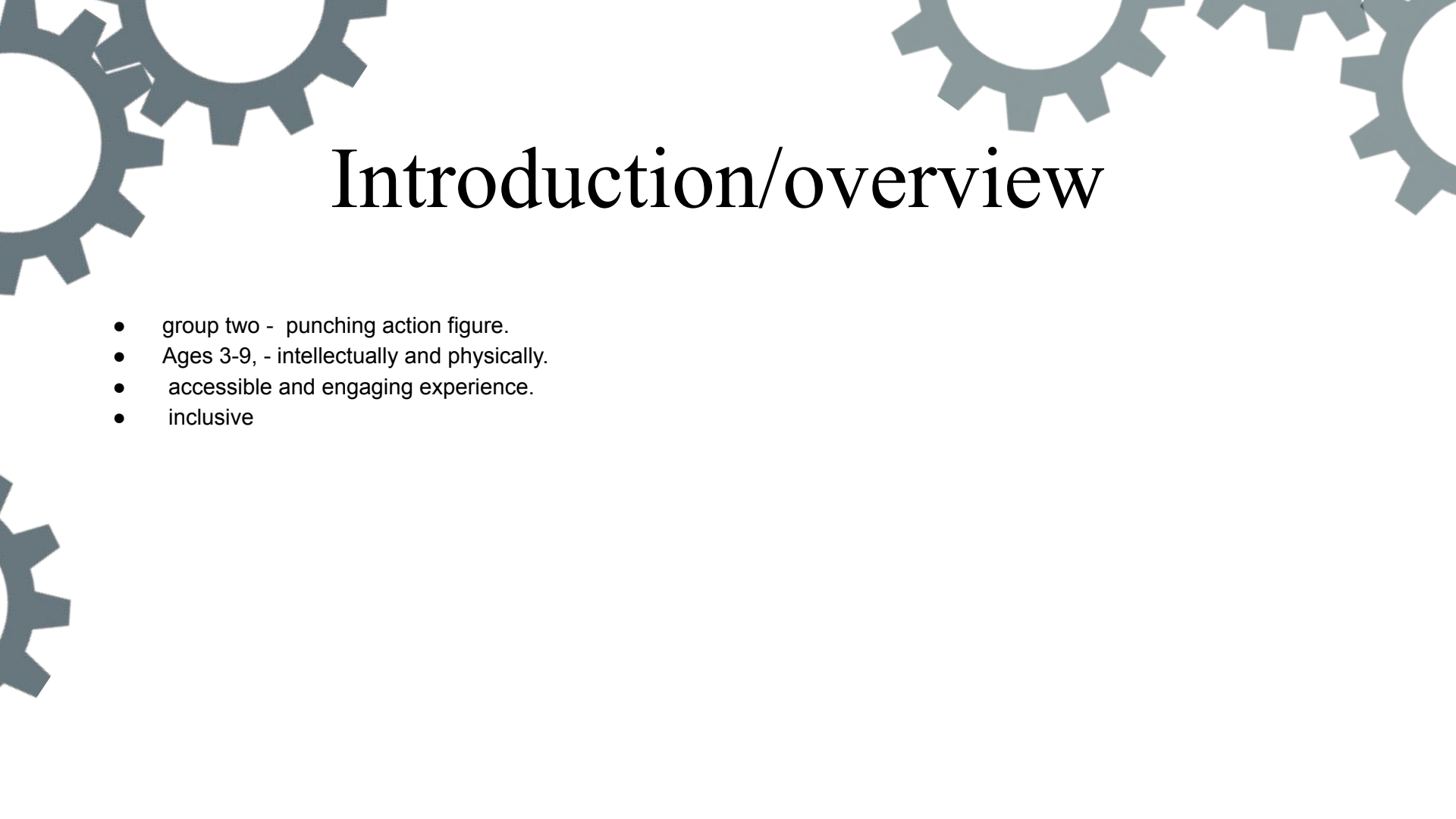




Universal Toy Design Project

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Group two



Introduction/overview

- group two - punching action figure.
- Ages 3-9, - intellectually and physically.
- accessible and engaging experience.
- inclusive

Background and Problem Statement

Why Our toy:

- **Importance:** Activity is crucial to a child's development all around especially the cognitive, social, and motor aspect of children's development. However, toys often fail to accommodate children with disabilities, leaving many excluded from these vital experiences.
- **Problem:** The toy market lacks affordable, engaging toys designed for children with diverse needs. Parents often face difficulty finding toys that their children can fully interact with
- For example, toys requiring fine motor skills or strength are inaccessible to children with physical limitations.
- **Impact:** This exclusion can lead to feelings of frustration, isolation, and missed developmental opportunities.

- **Accessible Mechanism:** The Punching Action Figure uses large, easy-to-press buttons and tactile grips that require minimal force, making it accessible for children with limited motor skills.
- **Inclusive Design:** It accommodates a wide range of abilities, allowing children with physical, cognitive, or sensory disabilities to fully engage with the toy. It is 4in x 4in x 6 in and for 3-9
- **Durable and Safe:** Constructed with lightweight, durable materials, the toy is designed for long-term use and safety, making it suitable for a variety of environments, from homes to therapy settings.



Design Solution Overview

What is our toy?

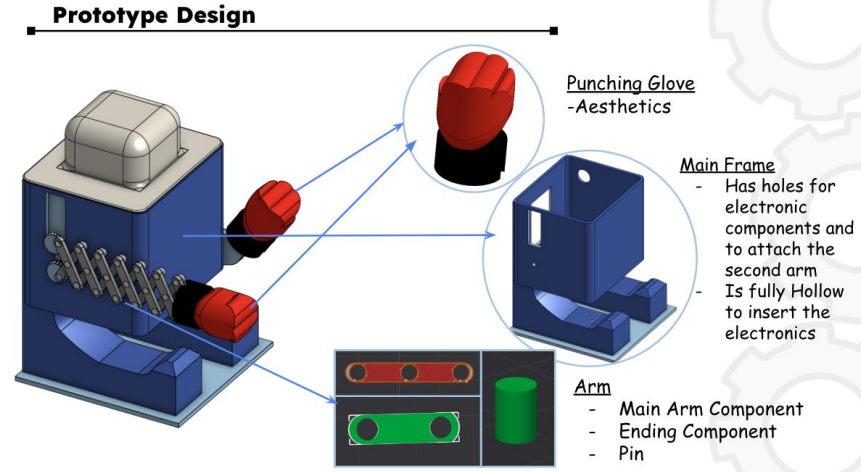
- interactive, user-friendly, accommodate

How it works:

- large, easy-to-press buttons, soft grips
- Spring mechanism, low power motor

Features that set it apart :

- Lightweight, durable, modular



Universal Design

Primary Benefit:

- The Punching Action Figure promotes inclusive play, allowing children with and without disabilities to engage together.

Key Accessibility Features:

- Large buttons and handles for ease of use.
- Adaptable for children with limited strength or motor control.
- Safe materials and simple mechanics to minimize risks.

Additional Advantages:

- Affordable manufacturing ensures accessibility for families of all socioeconomic backgrounds.
- The durable design ensures the toy withstands regular use, making it long-lasting and sustainable.

Broader Impact:

- This toy encourages physical activity, develops motor skills, and fosters social interaction in an inclusive environment. It is a step in reducing barriers to play for children with disabilities.



Challenges to Implications

Key Challenges:

- "Designing a toy that balances affordability with functionality while meeting the diverse needs of children with disabilities."
- "Ensuring the materials and mechanics are both safe and durable for a range of use cases."
- "Securing funding for mass production and widespread distribution."

How We Addressed These Challenges:

- The toy uses simple, cost-effective materials that are lightweight and durable.
- The design process involved consultation with parents and therapists to ensure it meets real-world needs.
- Learning from organizations like CP Family Network to ensure accessibility and affordability.

Successful Examples in Market

Inspiration From Existing Models: Zoomer Interactive Robots:

- Zoomer robots are interactive toys that respond to sound, touch, and movement. These toys incorporate motion-based interactivity, which is an inspiration for our Punching Action Figure, where physical touch (pressing buttons or grabbing) leads to action (punching movements or sounds).
- This may seem simple but it stimulates intellectual growth. Maybe not for adults but kids with the brain of a peanut can lead to stimulation and intellectual intelligence.
- We found this interesting as it can be useful for kids with every and any type of disability and is accessible for all

Similar Successful Designs:

- Toys like the switch-adapted remote control cars have shown that accessible designs can be highly engaging and widely used.
- Toys like these lead to small simulation, thinking and a bit of problem-solving
- These types of interaction are essential to grow skills from a young age to develop kids intellectually

Our Contribution:

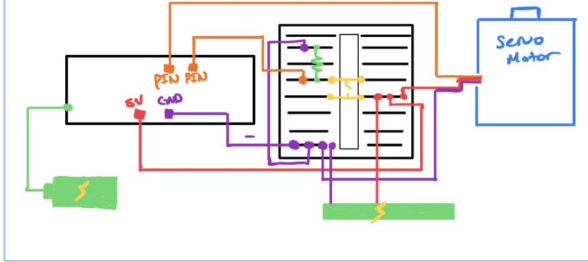
- The Punching Action Figure adds to this market by focusing on inclusive play for younger children, a group often overlooked in accessibility efforts.
- There has been a trend of kids being stuck on the phone or ipad. Otherwise, known as "ipad kids" these kids lack the social, motor and creative skills needed to be on a good development track.
- These kids lack the face to face interactions and social skills ending up harming communication skills, but also stay inside limiting motor and physical development. These are only a few of the negative sides
- We wanted to develop a toy where it is as stimulating and enjoyable as a ipad without the side effects of a screen in your face. Our toy will stimulate, develop and help your kid become intellectual smarter day by day.

Conclusion

1. Consumer Product Safety Commission. (2023). *Toy safety: ASTM F963 requirements*. U.S. Consumer Product Safety Commission. Retrieved from <https://www.cpsc.gov>
2. Cerebral Palsy Toy Network. (2024). *Toy modifications for children with cerebral palsy*. CP Family Network.
3. GutCheck. (2015). *Parents weigh in: Toy buying consumer research*. Retrieved from <https://gutcheckit.com>
4. KidPowered. (2022). *8 characteristics that make a toy great*. Retrieved from <https://kidpowered.com>
5. LanSchool. (2023). *Accessibility features for children's toys*. LanSchool. Retrieved from <https://lanschool.com>

Circuit Work

Group Project Wiring Diagram



Legend

- Ground
- Power
- Pins
- Resistor

- Two Pin Modes
 - Servo Program
 - Button Program
- Two batteries
- Control Board
- Additional Power to Servo

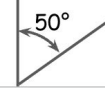
Description Of Code

```

1 #include <Servo.h>
2 //Declaration of Servo
3 Servo servo;
4 //Pin code for Button
5 int button = 3;
6 //Servo Angles
7 int servoMax = 180;
8 int servoMin = 0;
9 //Count for servo directions
10 int count = 0;
11 //Counts for Button Status Comparison
12 int p1 = 0;
13 int p2 = 0;
14 long distance;
15 void setup() {
16     //Testing Button Feedback
17     Serial.begin(9600);
18     //Initializing servo and button
19     servo.attach(2);
20     pinMode(button, INPUT_PULLUP);
21     pinMode(2, OUTPUT);
22 }
23
24 void loop() {
25     //Button Debounce
26     Serial.println(digitalRead(button));
27     //Configure Initial button state
28     digitalWrite(button, LOW);
29     //When Pressing button Code
30     if (p1 == HIGH && p2 == LOW && count == 0)
31     {
32         servo.write(servoMax);
33         count++;
34     }
35     //When no push on button code
36     if (digitalRead(button) == LOW && count == 1)
37     {
38         servo.write(servoMin);
39         count--;
40     }
41     //resetting count
42     if (count == 21)
43     {
44         count = 0;
45     }
46 }

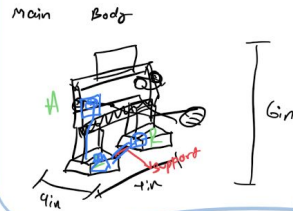
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- Initializing Variables:
 - Pins
 - Servo parameters
 - True False Counter
- Initialize Servo and PinMode
- If Statements
 - Checks previous setting
 - Increases count

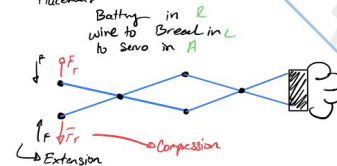


Initial Brainstorming

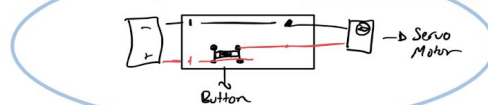
Group Design



Placement



Wiring



APA Citations

1. Consumer Product Safety Commission. (2023). *Toy safety: ASTM F963 requirements*. U.S. Consumer Product Safety Commission. Retrieved from <https://www.cpsc.gov>
2. Cerebral Palsy Toy Network. (2024). *Toy modifications for children with cerebral palsy*. CP Family Network.
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